ARTIFICIAL INTELLIGENCE

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AI Foundations

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What is AI?

- It is a branch of CS by which we can create an intelligent machine, which can behave like a human and think like a human and able to make decisions.

- Artificial---------”man made”
- Intelligence-------- the ability of making artificial thing----”thinking power”

- AI is the doorway of converting a simple computer into an intelligent computer. AI is based on how the human brain works, thinks, and decodes problems to arrive at solutions.

Why AI:

With the help of AI,

- We can create such amazing software's ---- a device which can solve real-world problems accurately and easily.

- We can create our personal virtual assistance.

- We can build such robots which can work in a environment where survival of human can be at a risk .

- AI opens path for new technologies, new devices and new opportunities

AI systems demonstrate capabilities

such as:

- Learning from experience : Netflix - recommendation
- Logical reasoning: Google Maps finds the best route
- Problem solving : A robot vacuum cleaner detects obstacles
- Decision making : Email spam filter decides whether an email is spam or should go to your inbox

Imagine a self-driving car:

Learning from Experience: Learns from millions of driving examples.

Logical Reasoning: Identifies traffic lights, pedestrians, and road signs.

Problem Solving: Finds another route if a road is blocked.

Decision Making: Decides whether to stop, slow down, or turn.

Exercise:

- A chess AI analyzes possible moves and chooses the best one.
- Face Unlock on a smartphone

John McCarthy   Father of Artificial Intelligence

American mathematician and computer scientist widely regarded as one of the founders of artificial intelligence.

He coined the term artificial intelligence in 1956.

helped establish AI as an academic discipline, and made foundational contributions to programming languages, computer systems, and machine reasoning.

Developed the programming language LISP, one of the earliest AI programming languages.

Artificial Intelligence is the science and engineering of making intelligent machines capable of performing tasks that normally require human intelligence. —

John McCarthy (1956)

History of AI

1950 – creating machines that think

1951 – Game AI – checkers program

1956 – John McCarthy coined the word AI in a conference

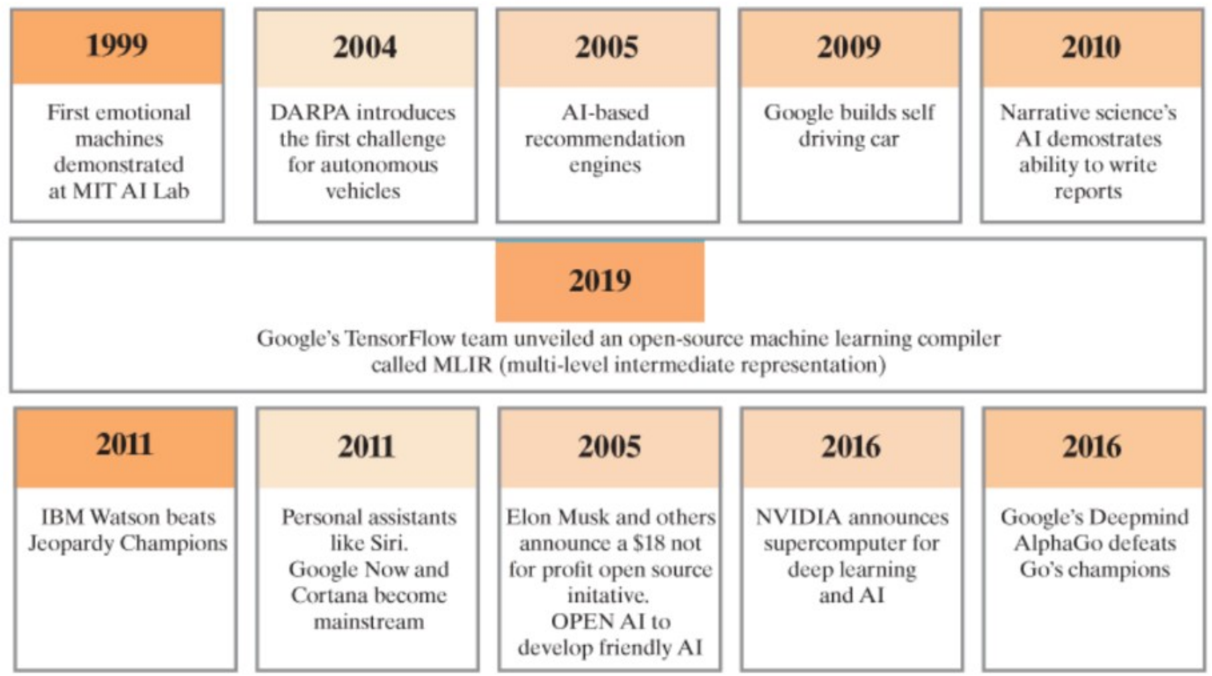
1961 – An AI chatbot ELIZA was introduced

2002 – “Roomba” Mass Produced Robotic Vacuum Cleaner

2014 – Amazon’s Alexa

2018 – NLP and Generative AI

2022 - ChatGPT



State-of-the-Art Technologies in Artificial Intelligence

most advanced, latest, and best technologies currently available in the field of Artificial Intelligence.

1.Autonomous planning and scheduling: **NASA's automatic planning application which had the ability to prepare the schedule of operating a spaceship**

2.Game strategizing: **IBM developed project Deep Blue, which became the first chess computer program to defeat the chess global champion, Garry Kasparov,**

3.Autonomous control system: **Autonomous Land Vehicle In a Neural Network (ALVINN) calculate the most optimal path to follow**

4. Medical diagnosis:

5. Robotics:

6.Natural language processing: chatbots.

Scope of AI

Discipline	Role in AI	Real-world Example
Computer Science	Programming and algorithms	Search engines, AI software
Mathematics	Logic, probability, optimization calculations and predictions	Fraud detection, weather prediction
Linguistics	Language understanding	ChatGPT, Google Translate
Psychology	Human learning and behavior	Recommendation systems
Neuroscience	Brain-inspired computing	Neural Networks, Deep Learning

Scope of AI

Robotics	Intelligent machines	Self-driving cars, drones
Healthcare	Medical diagnosis	AI-assisted disease detection
Finance	Intelligent financial analysis	Credit scoring, fraud detection
Cybersecurity	Threat detection	Spam filtering, malware detection
Generative AI	Content creation	ChatGPT, DALL·E, Gemini
AI Agents	Autonomous task execution	Virtual assistants, coding agents

APPLICATIONS OF ARTIFICIAL INTELLIGENCE

AI is transforming the world by solving real-world problems and enhancing human capabilities across diverse domains.

01 HEALTHCARE

AI helps in disease diagnosis, drug discovery, patient monitoring and personalized treatment.

Examples: IBM Watson, AI in Radiology

02 AUTOMOBILE

AI powers self-driving cars, traffic prediction, advanced driver assistance and safety systems.

Examples: Tesla Autopilot, Waymo

03 FINANCE

AI is used for fraud detection, algorithmic trading, risk assessment, and automated customer support.

Examples: Robo-Advisors, Fraud Detection

04 SURVEILLANCE

AI enables real-time monitoring, anomaly detection, face recognition and behavior analysis.

Examples: Smart CCTV, Face Recognition

05 SOCIAL MEDIA

AI personalizes content, detects fake news, analyzes user behavior and improves engagement.

Examples: Content Recommendation, Sentiment Analysis

06 ENTERTAINMENT

AI recommends movies, music and shows, creates special effects and enhances gaming experiences.

Examples: Netflix Recommendations, AI in Gaming

07 EDUCATION

AI personalizes learning, automates evaluation, and provides intelligent tutoring systems.

Examples: Duolingo, AI Tutors

08 SPACE EXPLORATION

AI analyzes space data, detects celestial objects and helps in mission planning.

Examples: NASA AI, Mars Rover

09 GAMING

AI creates intelligent NPCs, adapts game difficulty and enhances player experience.

Examples: AI Opponents, Game Design

10 ROBOTICS

AI enables robots to sense, learn, decide and perform tasks autonomously.

Examples: Industrial Robots, Drones

11 AGRICULTURE

AI helps in crop monitoring, yield prediction, pest detection and precision farming.

Examples: Crop Disease Detection, Smart Irrigation

12 E-COMMERCE

AI personalizes recommendations, manages inventory and improves customer service.

Examples: Amazon Recommendations, Chatbots

BENEFITS OF AI

- Increases Efficiency
- Better Decision Making
- 24/7 Availability
- Reduces Human Error
- Saves Time & Resources
- Drives Innovation

EMERGING AREAS OF AI

- Generative AI (ChatGPT, DALL-E)
- Natural Language Processing (NLP)
- Computer Vision
- AI Agents & Automation
- Edge AI & IoT
- Predictive Analytics

AI is not just a technology, it is the future of human innovation.

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AI vs Traditional Programming

```
TRADITIONAL PROGRAMMING

 Rules (Program) + Input Data
               |
               ▼
       Computer Program
               |
               ▼
            Output
```

```
ARTIFICIAL INTELLIGENCE

 Input Data + Correct Output
              |
              ▼
      Machine Learning
              |
              ▼
          AI Model
              |
        New Input Data
              |
              ▼
      Predicted Output
```

```
if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

```
Thousands of Cat Images
+
Thousands of Dog Images
        |
        ▼
 Train AI Model
        |
        ▼
New Image
        |
        ▼
Prediction:
🐱 Cat
or
🐶 Dog
```

Artificial Intelligence (AI) can be classified in two ways:

1. Based on Capability (Intelligence Level) : This classification describes how intelligent an AI system is and what level of tasks it can perform.

2. Based on Functionality (Working Principle) : This classification explains how AI systems work.

Types of AI

Artificial Intelligence (Intelligence Level)

- Narrow AI
- General AI
- Super AI

Artificial Intelligence(Working Principle)

- Reactive Machines
- Limited Memory
- Theory of Mind AI
- Self-Aware AI

Types of AI - Based on Capability

1. Artificial Narrow Intelligence (ANI)

- Also called as Weak AI
- Narrow AI is designed to perform one specific task. It cannot perform tasks beyond the purpose for which it is trained.
- **Characteristics**
- Performs only one specialized task.
- Cannot think like humans.
- Works within predefined limits.
- Most AI systems today belong to this category.

Examples

ChatGPT

Google Translate

Siri and Alexa

Face Recognition

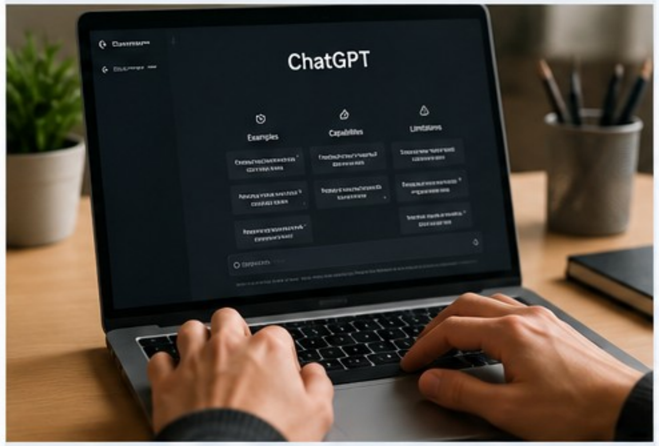
Netflix Recommendations

Spam Email Detection

Types of AI - Based on Capability

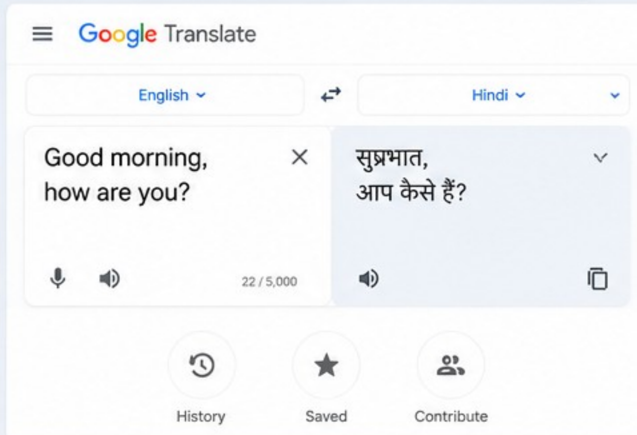
REAL-LIFE EXAMPLES OF AI (NARROW AI)

1. ChatGPT



ChatGPT is an AI chatbot that understands questions and generates human-like responses.

2. Google Translate

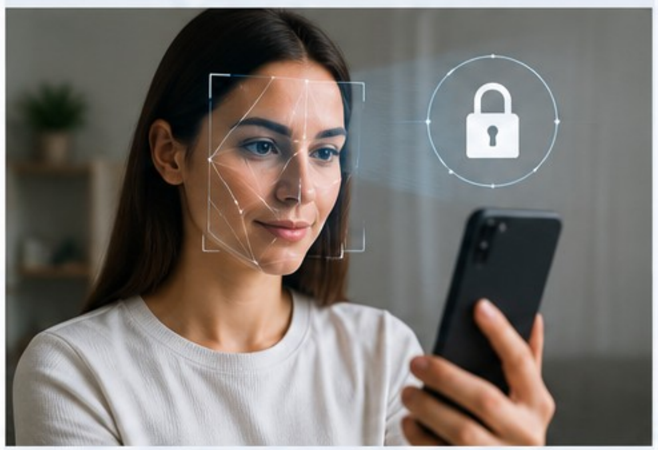


Google Translate uses AI to translate text and speech from one language to another.

3. Siri and Alexa

Siri and Alexa are AI virtual assistants that answer questions, set reminders, play music, etc.

4. Face Recognition

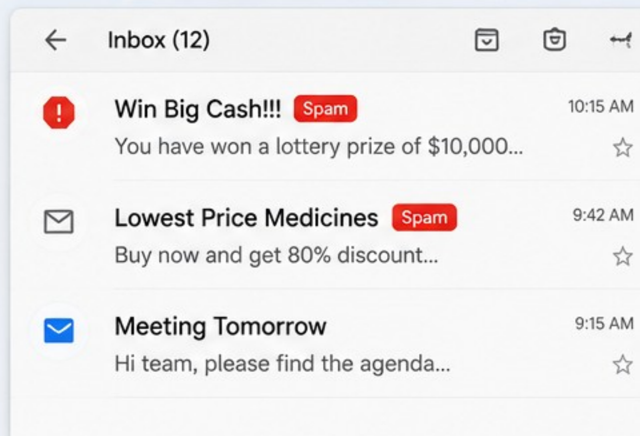


Face recognition AI is used in smartphone unlocking, security systems, and attendance.

5. Netflix Recommendations

Netflix uses AI to analyze your watching behavior and recommend shows and movies you may like.

6. Spam Email Detection



Email services use AI to detect spam messages and keep our inbox clean and safe.

Types of AI - Based on Capability

2. Artificial General Intelligence (AGI) -

- Also called as Strong AI
- General AI refers to machines that can perform any intellectual task that a human can perform.

Characteristics

Learns different types of tasks.

Thinks, reasons, and solves problems.

Can adapt to new situations.

Human-level intelligence.

Currently no true General AI exists.

Possible future examples:

Human-like robots

Intelligent personal assistants capable of performing any task

Still under research.

Extremely difficult to build.

Raises ethical and safety concerns.

Types of AI - Based on Capability

Types of AI - Based on Capability

3. Artificial Super Intelligence (ASI)

Capability of machines to surpass human beings. Super AI is an AI system whose intelligence exceeds that of humans in every field.

Characteristics

Smarter than humans.

Can solve highly complex problems.

Learns and improves independently.

Examples

Currently does not exist.
Examples found only in science fiction:
Hypothetical (imagined, assumed, or predicted)

Types of AI - Based on Capability

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1. Discovering New Laws of Physics

AI systems like DeepMind's AlphaFold predict protein structures, helping scientists discover new biological and physical principles.

2. Ending Global Poverty

AI helps optimize resource distribution, improve education, healthcare, and create equal opportunities for all.

3. Disaster Management

AI analyzes data from satellites, sensors, and social media to predict disasters early and coordinate faster rescue and relief operations.

4. Managing Extraterrestrial Exploration

AI-powered robots and systems explore other planets, analyze data, and communicate with astronauts for safe missions.

Types of AI : Based on Functionality

4 Types of AI – Based on Functionalities

1. **Reactive Machine AI**
2. **Limited Memory AI**
3. **Theory of Mind AI**
4. **Self-Aware AI**

Types of AI : Based on Functionality

1. Reactive Machine AI

- Reactive machines respond only to the current situation. They do not store memories or learn from previous experiences.
- Characteristics
- No memory.
- No learning.
- Responds only to current input.
- Simple decision making.
- Example
- IBM Deep Blue Chess Computer
- Real-Life Example
- A calculator performs calculations based only on the numbers entered.

Types of AI : Based on Functionality

Types of AI : Based on Functionality

2. Limited Memory AI

- Limited Memory AI uses past information together with current data to make decisions.
- Characteristics
- Stores temporary information.
- Learns from previous experiences.
- Most modern AI systems belong to this category.
- Examples
- Self-driving cars
- ChatGPT
- Google Maps
- Recommendation systems

3. Theory of Mind AI

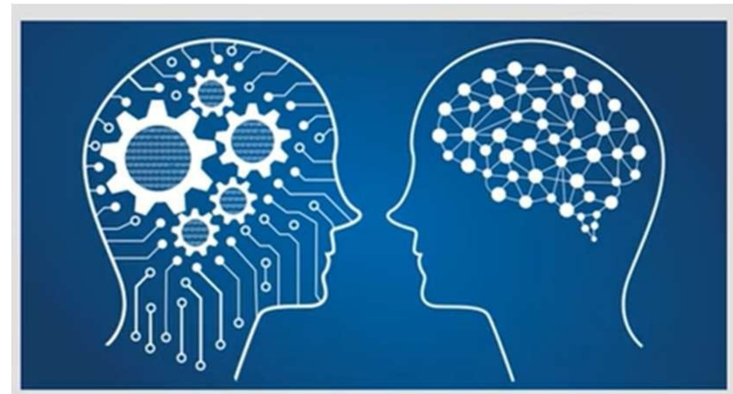
- Theory of Mind AI is expected to understand human emotions, beliefs, intentions, and behaviour.

Characteristics

Understands emotions.

Interacts naturally with humans.

Makes socially intelligent decisions.



Examples

Currently under research.

Possible future applications:

Emotional robots

AI caregivers

AI teachers

Types of AI : Based on Functionality

4. **Self-Aware AI**

- Self-aware AI is an AI system that possesses self-consciousness, emotions, and awareness of its own existence.

AI is the fundamental risk to the existence of human civilization – Elon Musk

Characteristics

Self-conscious.

Independent thinking.

Makes autonomous decisions.

Examples

Currently does not exist.

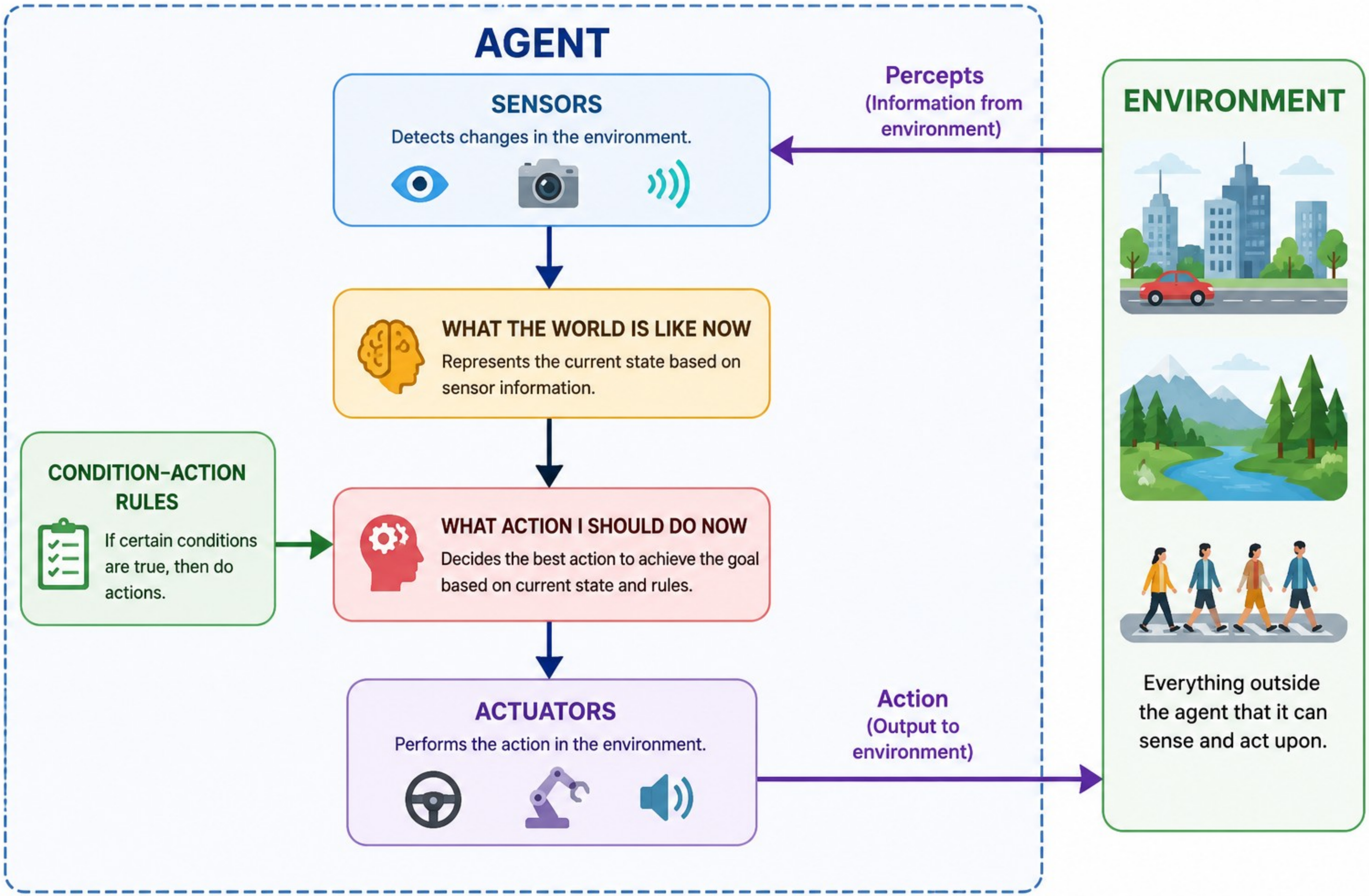
What is an Agent?

An Agent is an entity (software, robot, or machine) that perceives its environment through sensors and acts upon the environment through actuators to achieve a specific goal.

Environment
↓
Sensors
↓
Agent
↓
Actuators
↓
Environment

Component	Description
Environment	The surroundings in which the agent operates
Sensors	Collect information from the environment
Agent	Processes information and makes decisions
Actuators	Perform actions

Intelligent Agent



Intelligent Agent

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1. Example: A robot vacuum cleaner senses dust (sensor) and moves to clean (actuator).

1.2. Self-Driving Car

2.3. Chess AI

3.4. Google Maps

4.Autonomous Drone

5.Smart Home Assistant

6.Smart Traffic Signal

7.Face Unlock Agent

Write AI agenet component for the above AI System.

Characteristics

Autonomy: Operates without human intervention

Perception: Collects information using sensors

Action: Performs task using ia actuators

Goal-oriented: Acts to achieve specific objectives

Adaptability: May learn or adapt based on experience

Internal Working

An Agent Function defines what action should be taken for every possible percept sequence.

IF obstacle detected

↓

Stop

An Agent Program is the actual software (code) that implements the agent function.

```
if obstacle == True:
    Stop()
```

Preprocessors take the raw data from sensors and convert it into a usable format.

Input
|
Sensor
|
Preprocessor
|
Agent Function
|
Actuator
|
Action

Example: Self-Driving Car Agent

- Sensors: Cameras, radar, GPS
- Actuators: Steering, brakes
- Agent Program: Detects lanes, avoids obstacles, navigates

Types of Intelligent Agents

1. Simple Reflex Agent
2. Model-Based Reflex Agent
3. Goal-Based Agent
4. Utility-Based Agent
5. Learning Agent

Intelligent Agent

Type of Agent	Single-Statement Definition	Real-Time Example
Simple Reflex Agent	Acts only on the current input using predefined **if–then rules**, without considering past experiences.	Automatic sliding door Thermostat: Turns heating on/off based on current temperature
Model-Based Reflex Agent	Uses the current input along with an **internal memory (model)** of the environment to make decisions.	Robot vacuum cleaner Vacuum robot remembers areas it has already cleaned
Goal-Based Agent	Selects actions by evaluating different possibilities to **achieve a specific goal**.	Google Maps navigation

Intelligent Agent

Utility-Based Agent	Chooses the action that provides the **maximum benefit or best outcome** among several alternatives.	Self-driving car selecting the safest and fastest route Investment AI choosing between high-return and low-risk options
Learning Agent	Learns from **past experiences** and continuously improves its performance over time.	ChatGPT or Netflix recommendation system

Activity 1: Case Study

Question:

Consider the following AI systems. Identify the type of intelligent agent and explain your answer in one sentence.

Face Unlock on a smartphone
Google Assistant
Smart Traffic Signal
Spam Email Filter
Amazon Product Recommendation
Autonomous Delivery Robot

Activity 2: Think and Identify

Question:

Read each scenario and identify the type of intelligent agent.

A smart door opens automatically when a person approaches.

A robot remembers obstacles and avoids them while cleaning.

Google Maps finds the shortest route to your destination.

A self-driving car chooses the safest and fastest route.

ChatGPT improves its responses based on training and user interactions.

Activity 3: Real-World AI Challenge

Question:
Look around your home or college and identify five AI-based systems. Classify each one into the appropriate type of intelligent agent and justify your answer.

AI System	Agent Type	Justification
Example: Google Maps	Goal-Based Agent	Finds the shortest route to reach the destination.

Environments

Agents operate **within environments** — which shape their behavior and performance

An environment is everything surrounding an intelligent agent that it can perceive, interact with, and act upon while performing a task.

The environment is the world in which an AI agent operates.

Example:

For a vacuum robot, the environment is the floor layout, obstacles, and dust

Self-Driving Car: Everything around the car that influences its driving decisions. (Roads, traffic signals, pedestrians, other vehicles)

The chessboard and the opponent's moves.

PEAS Model

PEAS is a framework used to describe and design an intelligent agent by specifying its Performance Measure, Environment, Actuators, and Sensors.

P: Performance Measure

E: Environment

A: Actuators

S: Sensors

Performance Measure (P)

This defines how the agent's success is measured, such as the number of tasks completed, the time taken, or the accuracy of its actions.

Environment (E)

This refers to the surroundings in which the agent operates, including its physical or virtual context.

Actuators (A)

These are the mechanisms through which the agent interacts with and changes its environment, such as motors, speakers, or display screens

Sensors (S)

These are the tools used by the agent to perceive and gather information about its environment, such as cameras, microphones, or touch sensors

PEAS Model for Smart Traffic Signal

SMART TRAFFIC SIGNAL

P → Reduce congestion & waiting time

E → Roads, Vehicles, Pedestrians

A → Traffic Lights, Display Boards

S → Cameras, Vehicle Sensors

PEAS Model for Chess-Playing AI

CHESS-PLAYING AI

P → Win the game

E → Chessboard, Opponent

A → Move Chess Piece

S → Board Position, Opponent's Move

Environment Types or Characteristics

1. Fully or Partially Observable
2. Deterministic vs Stochastic
3. Episodic vs Sequential
4. Static vs Dynamic
5. Discrete vs Continuous
6. Single-Agent vs Multi-Agent

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1. Fully or Partially Observable

Fully Observable

Agent has access to the complete state of the environment at each moment, before making a decision.

Example:

The AI can see the entire chessboard and all pieces.

Tic-Tac-Toe: The AI can see all 9 cells of the board.

Partially Observable

- The agent has incomplete or limited information about the environment. Some information is hidden or unavailable.
- Example: Driving in fog, blind spots, and hidden vehicles limit visibility.
- The AI makes best decisions using limited information from cameras, radar, and sensors.

Environments

2. Deterministic Vs Stochastic

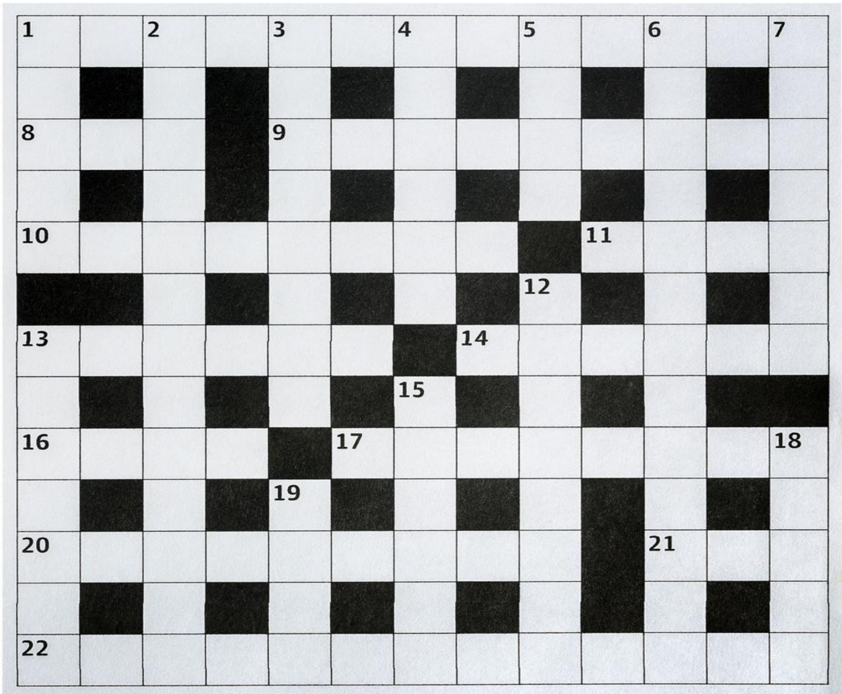
Deterministic

- Next state is completely determined by the current state and action. If the same action is repeated, it always produces the same result.
- Example: Crossword puzzle, Calculator
-

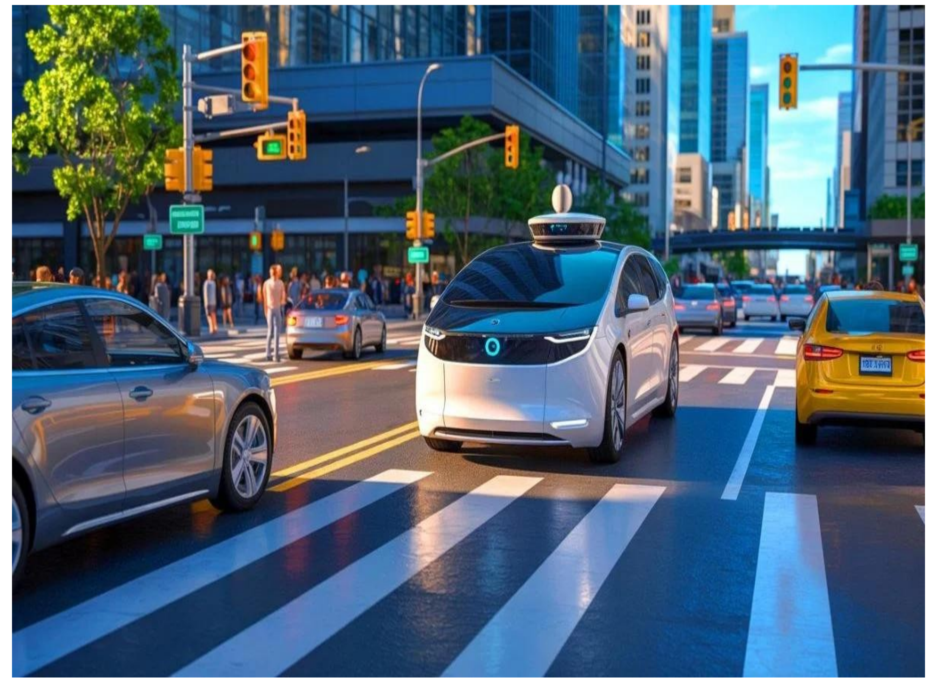
Stochastic

- The outcome is uncertain and may vary due to randomness.
- Example: Rolling a dice, stock market, Weather Forecasting AI

Environments



If the clue is 'Capital of India', the correct answer is New Delhi.
The answer does not change.



The car may choose to turn left.

A pedestrian suddenly crosses the road.

The AI must react to unexpected events.

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3. Episodic Vs Sequential

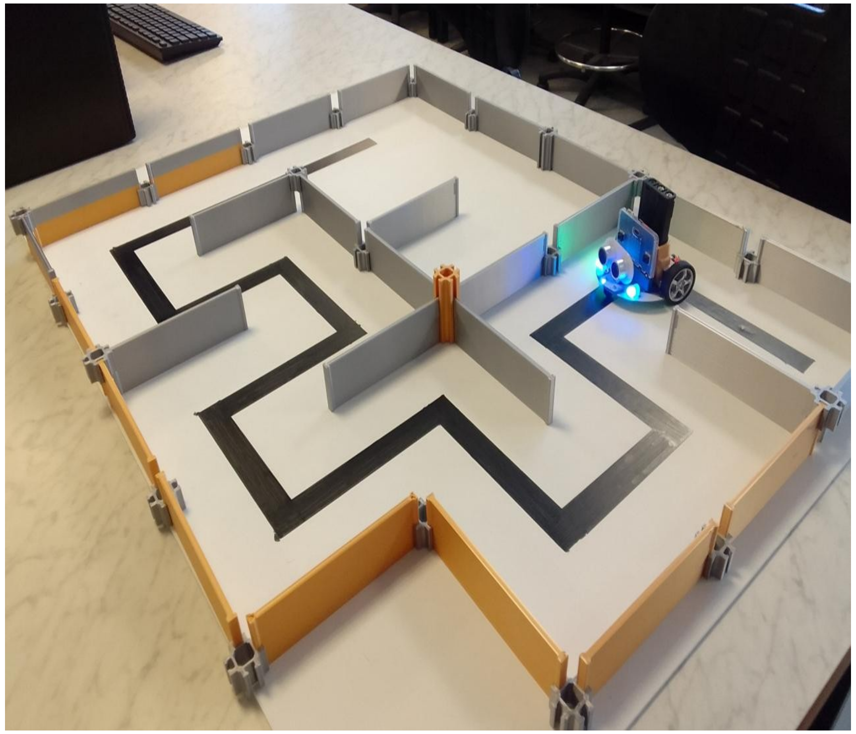
Episodic

- Each action is independent of previous actions. Decisions are based only on the current input.
- Example: Image classification, It classifies it as Cat, Dog, Car, etc. The next image is classified independently.
- Face Recognition at Airport : Each passenger is identified independently. The AI checks whether the face matches the passport.

Sequential

- Current decisions affect future actions. decisions. The AI must remember previous actions because they influence what happens next.
- Example: Chess, Driving : If the car changes lanes now, it affects the next few seconds of driving. .
- Every movement affects the next path and final delivery.

Environments



Robot Navigating a Maze
The robot must remember where it has already traveled.

4. Single-Agent Vs Multi-Agent

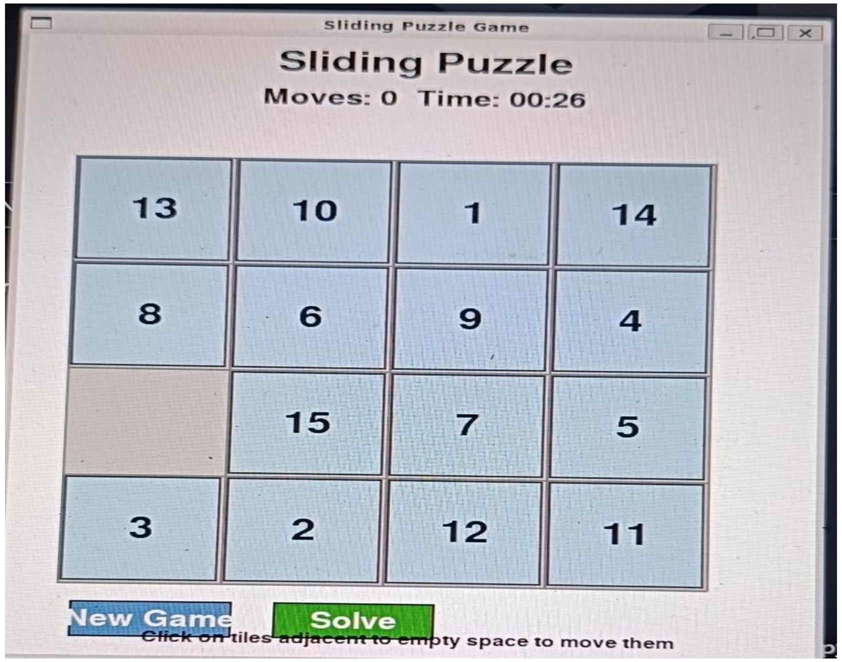
Single-Agent

- where only one AI agent operates alone. Its success depends only on its own actions, not on other agents.
- Example: Puzzle-solving bot, Robot Vacuum Cleaner

Multi-Agent

- where two or more agents interact, cooperate, or compete. The actions of one agent can affect the decisions of the others.
- Warehouse Robots (Cooperation) : Multiple robots work together to move packages.
- autonomous cars on road:Each car observes nearby vehicles and adjusts its speed or direction. Each player's move affects the other player's strategy.

Environments

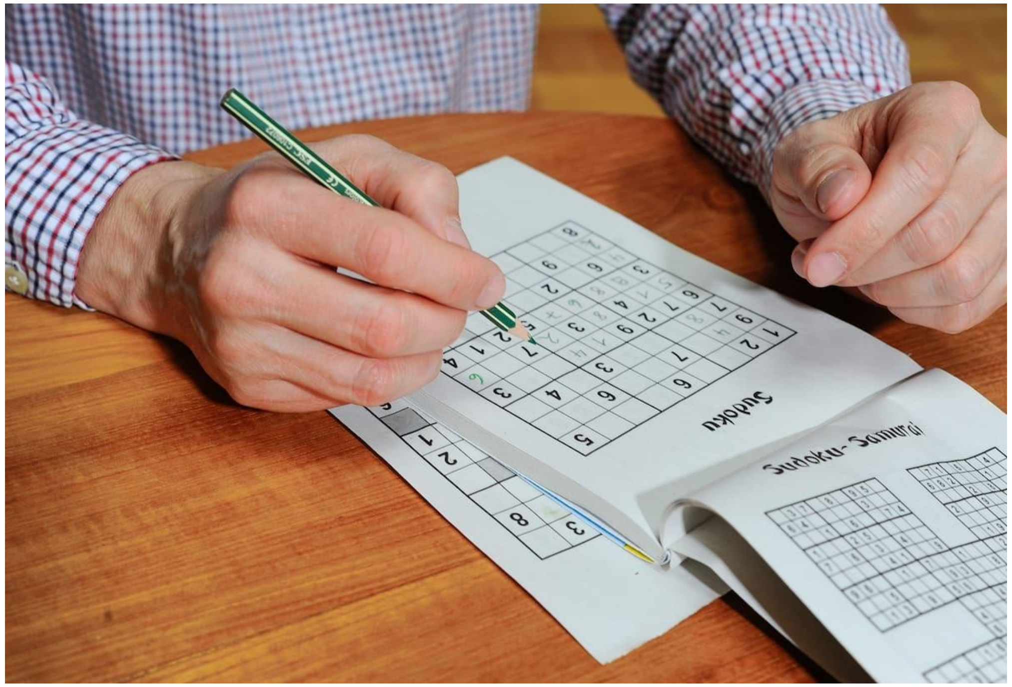


5. Static Vs Dynamic

Static

- The environment does not change while the agent is making a decision.
- Example: Sudoku Solver : No numbers change automatically. The AI can think as long as needed.
- Crossword Puzzle: The clues and empty boxes do not change.
- **Dynamic**
- where the environment changes continuously while the AI is making a decision.
- Example: Smart Traffic Management System
- Vehicles are constantly moving. Traffic lights must adapt to changing traffic conditions.
- The AI continuously updates its decisions.

Environments



Service Robot in a Shopping Mall

People are walking in different directions.

The robot must avoid moving people and obstacles.

It keeps updating its path in real time.

6. Discrete Vs Continuous

Discrete

- The environment has a finite number of states or actions.
- Example: S Tic-Tac-Toe Game : Limited cells and possible moves

Continuous

- States and actions change continuously.
- Example: S Smartwatch Health Monitoring : Heart rate and body temperature change continuously.

Agent Effectiveness

Agent effectiveness refers to how well an intelligent agent performs its assigned task and achieves its objectives.

Agent's effectiveness is measured using:

- Performance Measures (PEAS) : Did the agent complete the task successfully?
- Goal Achievement : Did the agent achieve its goal?
- Learning & Adaptability: Can the agent improve from experience?
- Efficiency and Robustness : Did the agent complete the task using minimum time, cost, and resources?

PEAS Model for Driverless Cars

Performance Measure: The measure for driverless cars is safe navigation and efficient route planning, ensuring passenger safety and timely arrivals.

Environment: The environment includes roads, traffic patterns, pedestrians, and weather conditions, which the car must interact with while navigating.

Actuators: Actuators consist of steering, acceleration, and braking systems that execute the car's movements as directed by its AI algorithms.

Sensors: Sensors, such as cameras, LiDAR, GPS, and radar, collect real-time data about the car's surroundings

Environments - Activity

1. PEAS Model – Smart Watch (Health Monitoring)

PEAS	Description
Performance Measure	Accurately monitor heart rate, detect health issues, provide timely alerts
Environment	Human body, daily activities, surroundings
Actuators	Display screen, vibration, speaker, notification alerts
Sensors	Heart-rate sensor, SpO_2 ser gyroscope, GPS

2. PEAS Model – Smart Agriculture System

PEAS	Description
Performance Measure	Increase crop yield, reduce water usage, detect diseases early
Environment	Farm, soil, crops, weather
Actuators	Irrigation pumps, fertilizer sprayers, drones
Sensors	Soil moisture sensors, temperature sensors, humidity sensors, cameras

AI Application: Automatically irrigates crops only when soil moisture is low.

Environments - Activity

5. PEAS Model – Smart Home

PEAS	Description
Performance Measure	Improve comfort, security, and energy efficiency
Environment	Home
Actuators	Lights, fans, AC, door locks
Sensors	Motion sensors, temperature sensors, cameras

8. PEAS Model – AI-Based Online Shopping

PEAS	Description
Performance Measure	Recommend relevant products and improve customer satisfaction
Environment	E-commerce website
Actuators	Product recommendations, notifications
Sensors	Browsing history, search history, purchase history

Environments - Activity

7. PEAS Model – Smart Parking System

PEAS	Description
Performance Measure	Efficiently allocate parking spaces
Environment	Parking lot
Actuators	Digital display boards, entry gates
Sensors	Ultrasonic sensors, cameras

10. PEAS Model – AI Classroom Attendance

PEAS	Description
Performance Measure	Accurate attendance with minimal errors
Environment	Classroom
Actuators	Attendance system, notifications
Sensors	Face recognition camera, RFID reader

Environments - Activity

System	Agent	Environment
ChatGPT	ChatGPT AI	User's questions and conversation
Amazon Alexa	Alexa AI	Home environment, user voice commands
Netflix Recommendation	Recommendation AI	User watch history and available movies
Face Recognition System	Face Recognition AI	Human faces, camera input, lighting conditions
Spam Email Filter	Spam Detection AI	Incoming emails and email content

Identify environment and agent

For solving a problem, we first need to represent the problem, find its output repeatedly until we get an optimal solution, and then return the solution.

Now, there are a number of variables and terms that affect the transition from the start of the problem to reach the goal state or desired state.

What is Problem Solving in AI?

- A problem is a scenario where an agent starts at an initial state, has a goal, and must find a path to reach the goal.

State Space Search is a problem-solving technique in Artificial Intelligence where all possible states (situations) of a problem are explored to find a path from the initial state to the goal state.

Why is State Space Search Needed?

Many AI problems have multiple possible solutions.

The AI agent needs to explore different states and choose the best path to reach the goal.

Search is the process of navigating the problem space to find this path.

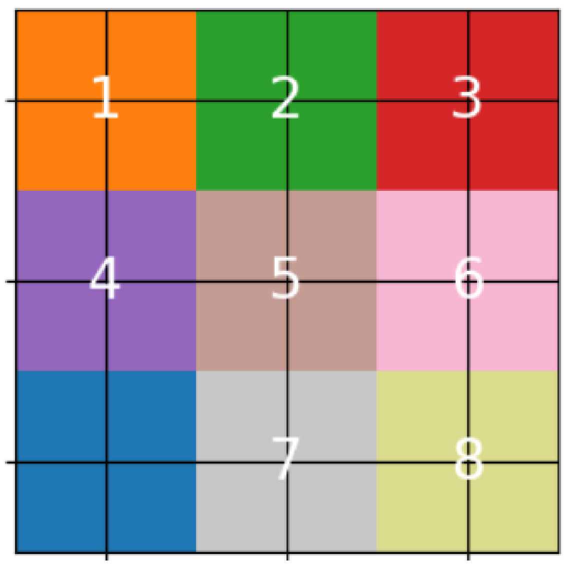
Why Search?

- AI agents often need to make sequences of decisions.
- Many AI problems can be formulated as search problems.

- Examples: Solving a puzzle, Route planning, Scheduling tasks, Game playing etc.

State

- A state is a specific configuration of the environment at a given moment.
-
- **Example-1: 8-Puzzle**



How AI Solves It

Observe the current puzzle.

Find all possible moves (up, down, left, right).

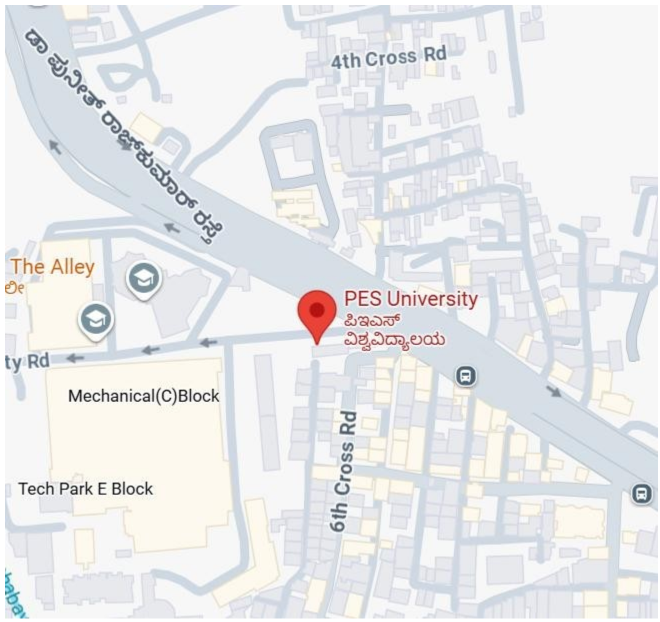
Create new puzzle states.

Continue searching until the correct arrangement is found.

State

- A state is a specific configuration of the environment at a given moment.

Example -2: Google maps



How AI Solves It

Finds all possible routes.
Calculates distance and travel time.
Compares traffic conditions.
Chooses the shortest or fastest route.
Goal: Reach the destination efficiently.

Search Strategies :

A search strategy determines how an AI searches for the best solution to a problem.

Types of Search Strategies

1. Uninformed Search

- No domain knowledge, Blind search; uses only problem definition
- Examples: Breadth-First Search (BFS) , Depth-First Search (DFS) , Uniform Cost Search (UCS)

2. Informed Search

- Uses heuristics (domain knowledge)
- Examples: A*, Greedy Best-First Search

Components of a State space search

- **State:** A specific configuration of the problem. A situation or condition of the problem.
- **State Space:** The collection of all possible states.
- **Initial State:** Starting point of the search
- **Goal State:** Desired end configuration
- **Transition:** An action that changes one state to another
- **Path:** A sequence of states connected by transitions
- **Search Strategy:** Method used to explore the state space

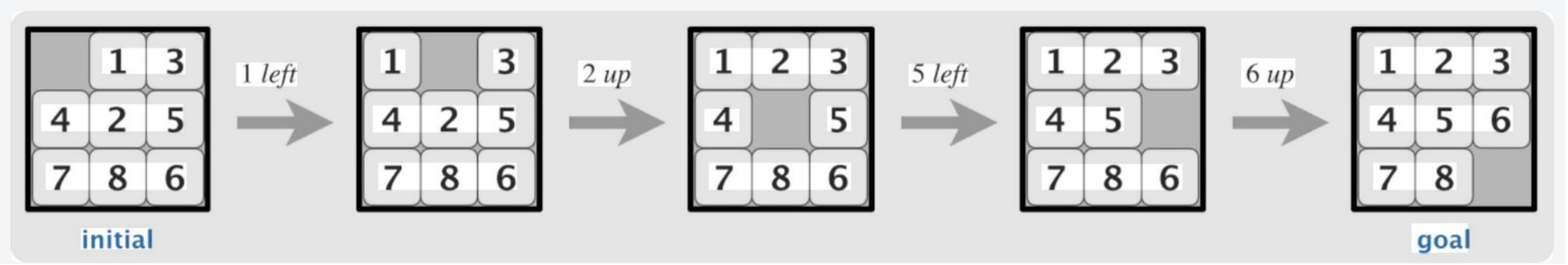
A State Space Tree is a tree representation of all possible states and actions that an AI agent can explore to reach the goal state.

```
        Start (S)
       /        \
   Right        Down
    |             |
 Dead End     Continue
                |
              Left     Right
               |         |
          Dead End     Goal
```

Root Node → Initial State (Start)
Branches → Possible action / direction (Up, Down, Left, Right)
Nodes → Different states
Leaf Node → Dead end or Goal
Every new position becomes a new state.
These states form a State Space Tree.
The search continues until the Goal is reached.

Example : 8 Puzzle Problem

- Initial State: A 3x3 tile board in any configuration
- Actions: Slide a tile up, down, left, or right
- Goal State: Reach a specific tile configuration
- Path Cost: Number of move



Example: Route Finding

Initial State: City A

Goal State: City B

Actions: Drive to neighboring cities

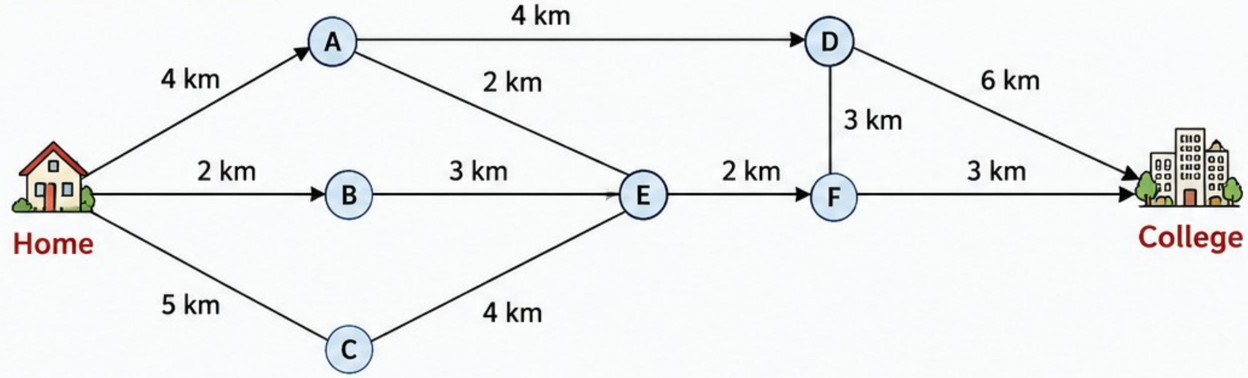
Transition Model: Distance/time between cities

Cost Function: Total distance or travel time

PES UNIVERSITY
CELEBRATING 50 YEARS

ROUTE FINDING IN STATE SPACE SEARCH

MAP (PROBLEM)



Problem
- Find the shortest route from Home to College.

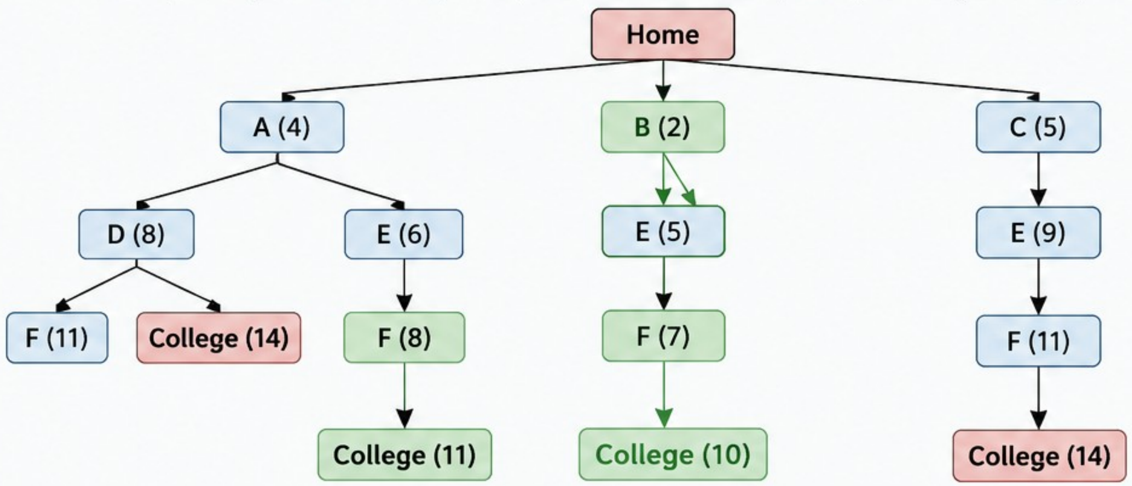
States
- Each city (node) is a state.

Actions
- Move from one city to a connected city.

Goal State
- College

STATE SPACE SEARCH

The search explores possible routes (states) from the initial state (Home) to the goal state (College).



Path Found (Optimal)

Home → B → E → F → College

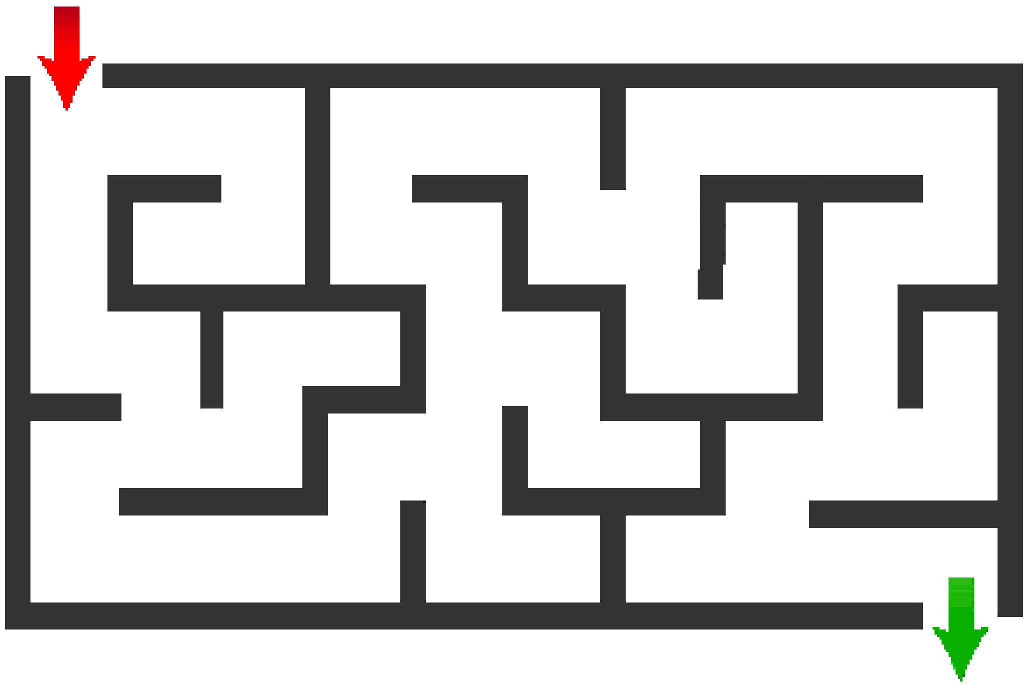
Total Cost = 10 km

Key Points
- Each node is a state.
- Numbers in brackets show total cost from start.
- Green path is the shortest route found.

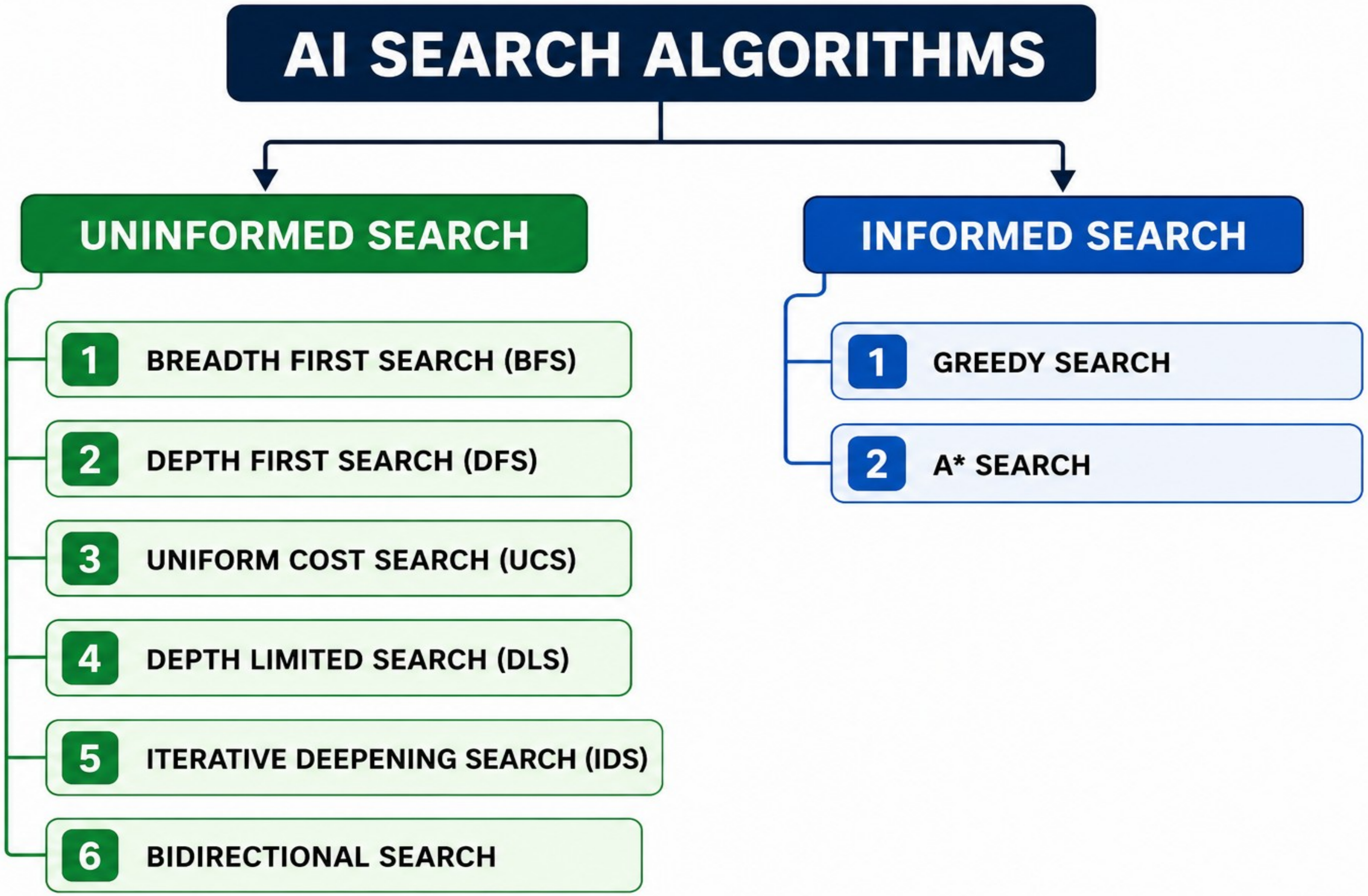
Explored States
Best Path
Other (Longer) Paths

Maze Problem in State Space Search

The Maze Problem is a classical AI problem in which an agent must find a path from the Start position to the Goal while avoiding obstacles (walls).



Uninformed Search



Uninformed Search

- Also called Blind Search
- No domain-specific knowledge — only the problem definition
- explores the search space without using any additional information (heuristics) about the goal state.
- Evaluates nodes based solely on their position in the search tree
- It doesn't use knowledge for the searching process
- It finds solution slow as compared to an informed search

Uninformed Search

- It is always complete
- Cost is high
- It consumes moderate time because of slow searching
- It is more lengthy while implemented
- It is comparatively less efficient as incurred cost is more and the speed of finding the Breadth-First solution is slow
- higher computational requirements
- Solving a massive search task is challenging
- Example Algorithms: Depth First Search (DFS), Breadth First Search (BFS)

Properties

- Completeness: Will it always find a solution?
- Optimality: Will it find the best (lowest cost) solution?
- Time complexity: Number of nodes generated
- Space complexity: Number of nodes stored in memory

BFS

- Breadth-First Search (BFS) is an uninformed search algorithm that explores the search space level by level. It visits all nodes at the current depth before moving to the next depth level.
- Uses a FIFO queue
- Complete and Optimal (if path cost is uniform)

Algorithm

- Initialize the frontier with the start node. (Insert it into the queue.)
- Loop:
  - Remove the first node from the frontier (queue)
  - If it's the goal → return solution
  - Else, expand it and add all unvisited successors to the end of the frontier (queue)
- Repeat until goal found or frontier is empty

```
        A
      / | \
     B  C  D
    / \    / \
   E   F  G   H
```

BFS Traversal

```
Level 0 : A
Level 1 : B  C  D
Level 2 : E  F  G  H
```

Order of traversal:

A → B → C → D → E → F → G

Robot Navigation

Explanation:

In robotics, BFS is used to find the shortest path from the robot's current position to its destination when all movements have equal cost. The robot explores all nearby locations first before moving farther away, ensuring it reaches the target using the minimum number of steps.

Example:

Consider a warehouse where an autonomous robot is assigned the task of picking up a package.

The robot knows the layout of the warehouse but must determine the shortest path from its current location to the package.

- BFS treats the warehouse as a graph, where:
- Nodes represent locations in the warehouse.
- Edges represent paths connecting those locations.

```
      A
     / \
    B   C
   / \   \
  D   E   F
       \
        G (Package)
```

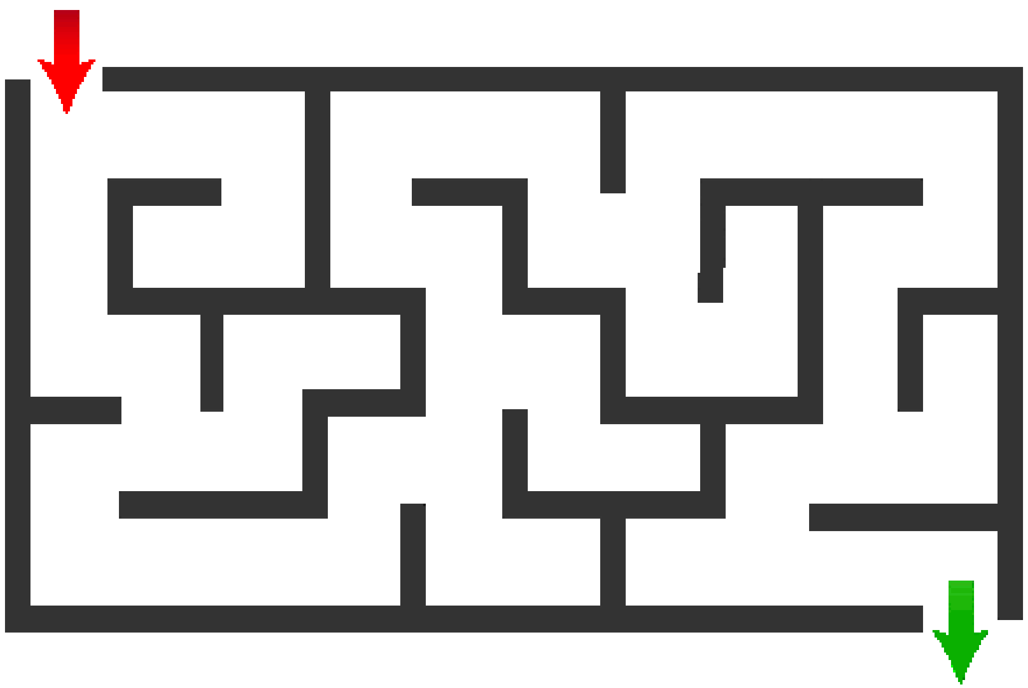
```
BFS Exploration Order:
Level 0 : A
Level 1 : B, C
Level 2 : D, E, F
Level 3 : G
```

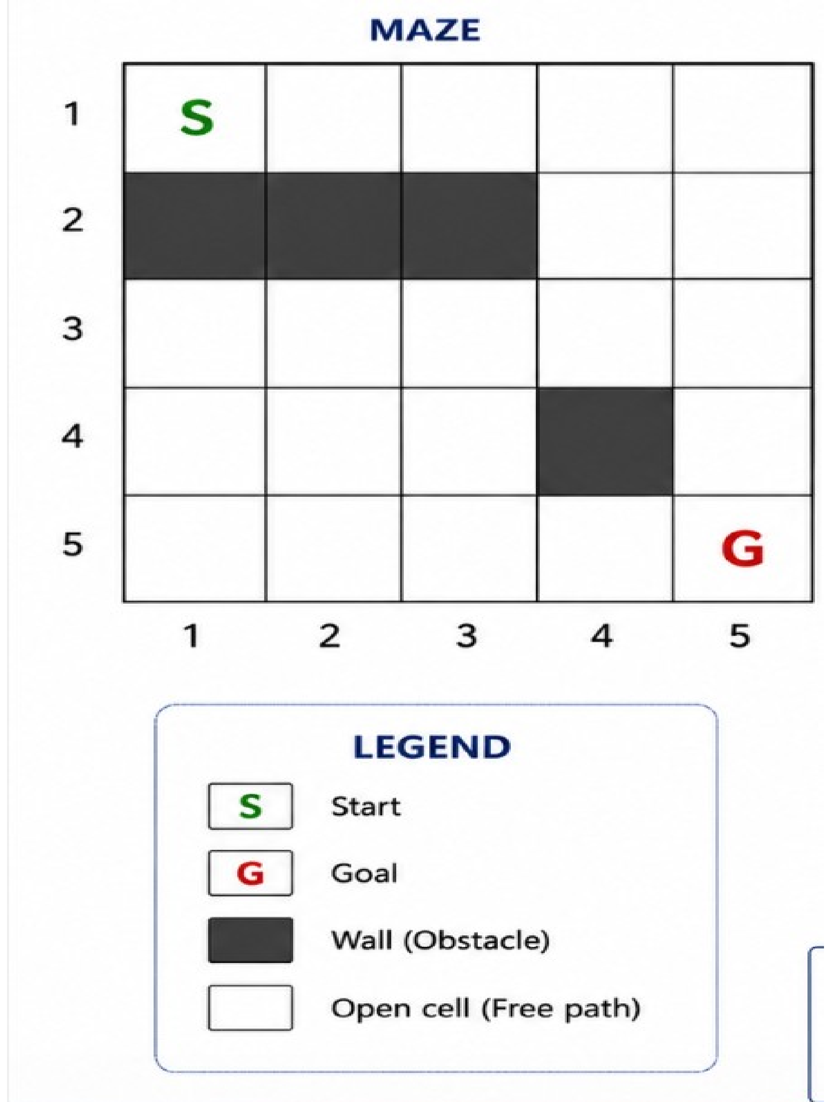
When it reaches G, it has found the shortest path:

A → B → E → G

Maze Solving

Imagine an AI-controlled robot or game character placed at the entrance of a maze. Its objective is to find the exit using the shortest possible path.





Consider the following social network, where each node represents a person and each connection represents a friendship.
Suppose Alice wants to find Grace using Breadth-First Search (BFS).

Questions

1)List the order in which BFS visits the people.

2)Which person is found first at each friendship level?

3)Write the shortest friendship path from Alice to Grace.

```
          Alice
         /     \
      Bob       Charlie
     /   \           \
 David   Emma         Frank
           \
           Grace
```

Answer:

BFS Traversal Order

`Alice → Bob → Charlie → David → Emma → Frank → Grace`

Shortest Friendship Path

`Alice → Bob → Emma → Grace`

If Alice wants to find Frank instead of Grace, will BFS still explore people level by level? What will be the traversal order and shortest path?

```
          Alice
         /     \
      Bob       Charlie
     /   \           \
 David   Emma         Frank
           \
           Grace
```

Informed Search

- Informed Search, also called Heuristic Search, is a search strategy that uses additional knowledge (heuristic information) to estimate which path is most likely to lead to the goal.

- This helps the algorithm reach the goal faster and more efficiently than uninformed search.

Simple analogy:

Imagine you are travelling from Bangaluru to Mysuru using Google Maps.

Uninformed Search checks many possible roads without knowing which one is closer to the destination.

Informed Search uses information such as distance, traffic, and estimated travel time to choose the most promising route.

Characteristics

Uses heuristic function (h(n)).
Searches toward the goal intelligently.
Explores fewer nodes.
Faster than uninformed search.
Suitable for large and complex search spaces.

Informed Search

- There is a direction given about the solution
- It is more efficient as it takes into account the cost and performance
- Computational requirements are lessened
- Having a wide scope in terms of handling large search problems

- Example: Greedy Search, A* Search, AO* Search, Hill Climbing Algorithm

Heuristic Function h(n)

- Definition: Estimates the cost from current node n to goal
- Example: In 8-puzzle → h(n) = number of misplaced tiles
- count how many numbered tiles are not in their correct position compared to the goal state.

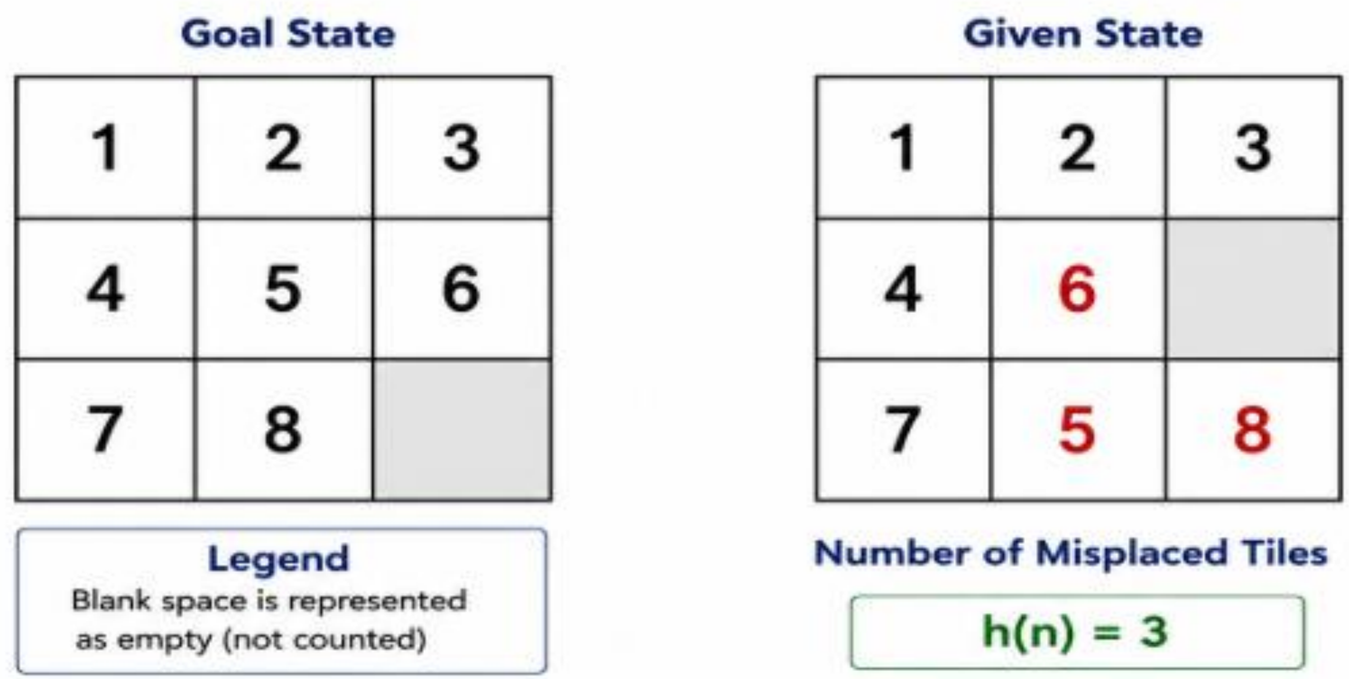
The AI calculates h(n) for every possible move.

- Smaller h(n) means closer to the goal.
- Larger h(n) means farther from the goal.

The AI prefers the state with the smallest heuristic value, because it appears to be the most promising path.

Characteristics

- Heuristic Function
- Efficiency
- Optimality and completenes



Comparison with Goal State

Tile	Goal Position	Given Position	Correct / Misplaced
1	(1,1)	(1,1)	Correct
2	(1,2)	(1,2)	Correct
3	(1,3)	(1,3)	Correct
4	(2,1)	(2,1)	Correct
5	(2,2)	(3,2)	Misplaced
6	(2,3)	(2,2)	Misplaced
7	(3,1)	(3,1)	Correct
8	(3,2)	(3,3)	Misplaced
Blank	(3,3)	(2,3)	Not Counted

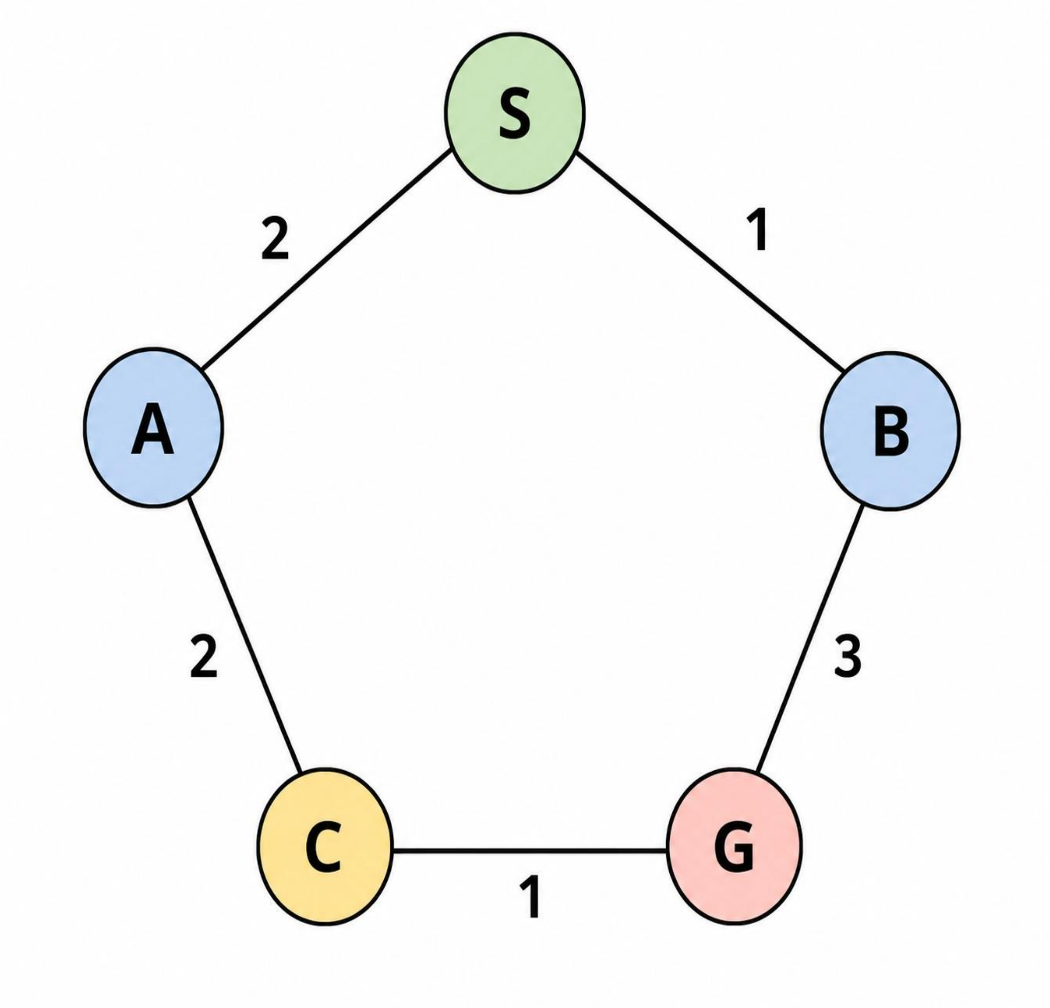
A* Algorithm

A* Search is an informed search algorithm that selects the path with the lowest total estimated cost using the evaluation function f(n) = g(n) + h(n)

It is complete and optimal when the heuristic is admissible.

- f(n)= g(n)+ h(n)
- f(n) = Total estimated cost
- g(n) = Actual cost from the Start node to node n
- h(n) = Estimated cost (heuristic) from node n to the Goal

Example 1



robot can move to either A or B.

The actual costs are:

S → A = 2
S → B = 1
A → C = 2
B → G = 3
C → G = 1

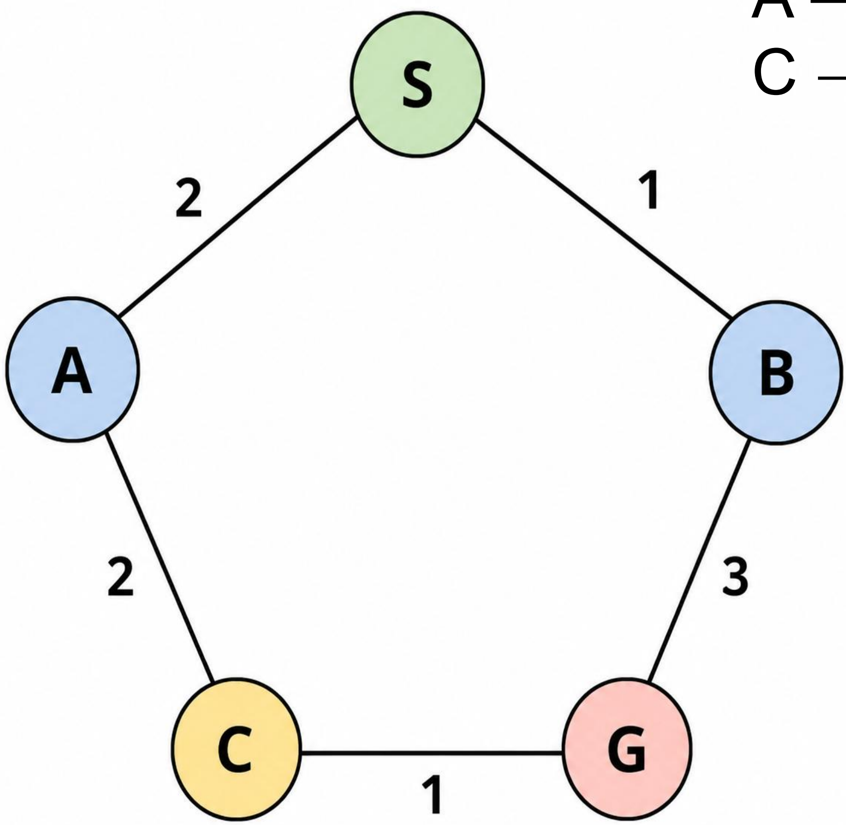
From Node A

actual cost from A to G is:

$S \rightarrow A = 2$

$A \rightarrow C = 2$

$C \rightarrow G = 1$



Actual cost from Start:

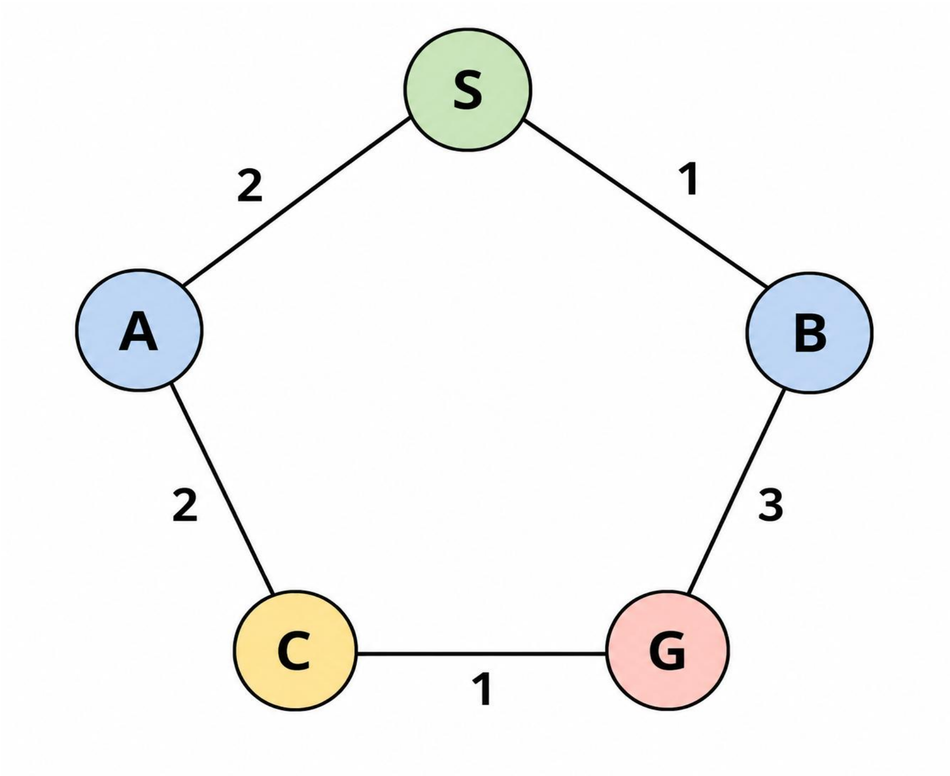
g(A) = 2

Estimated cost to Goal:

h(A) = 3

Total cost:

$$f(A) = g(A) + h(A) = 2 + 3 = 5$$



From Node B.

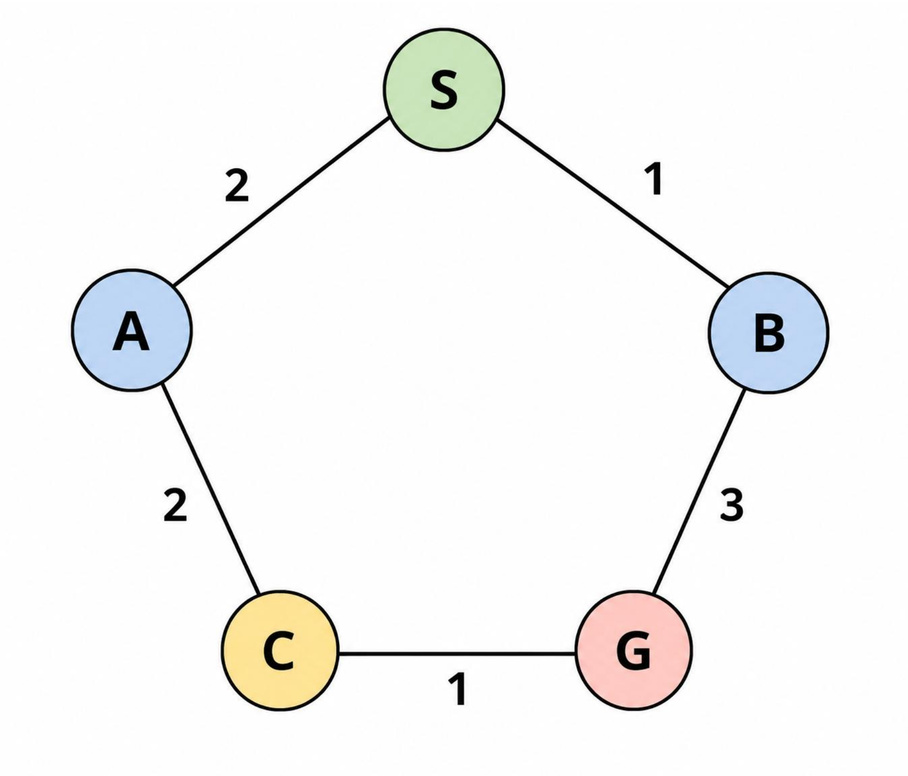
Actual cost from Start:
$g(B) = 1$

Estimated cost to Goal:
$h(B) = 3$

Total cost:
$$f(B) = g(B) + h(B) = 1 + 3 = 4$$

h(n) is the estimated cost (or estimated distance) from the current node to the goal node.

It is an estimate, not the exact cost.



Then show how A* calculates:
$f(A) = 2 + 3 = 5$
$f(B) = 1 + 3 = 4$
Since $4 < 5$, the algorithm chooses B.

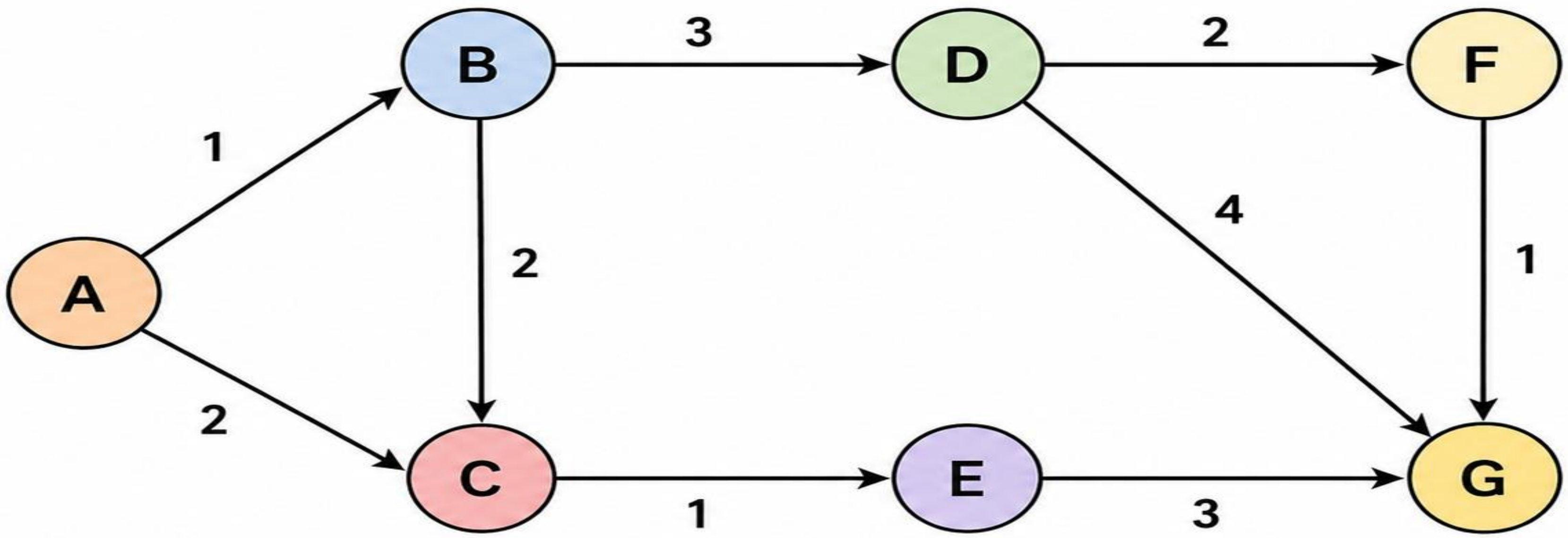
In real AI applications, the heuristic is designed according to the problem

for example, straight-line distance in GPS navigation or Manhattan distance in a grid.

Example 2

These values are assumed heuristic estimates.

Graph with Edge Costs (Updated Values)



Heuristic Values h(n)

Node	h(n)
A	5
B	6
C	4
D	4
E	3
F	1
G	0

Edge Costs:

- A → B = 1
- A → C = 2
- B → C = 2
- B → D = 3
- C → E = 1
- D → F = 2
- D → G = 4
- F → G = 1
- E → G = 3

Example 2

Step 1

A → B

$$g(B) = 1$$

$$h(B) = 3 + 4 = 7$$

$$f(B) = 1 + 7 = \boxed{8}$$

Step 2

A → C

Path from C to G:

$$C \rightarrow E \rightarrow G$$

$$g(C) = 2$$

$$h(C) = 1 + 3 = 4$$

$$f(C) = 2 + 4 = \boxed{6}$$

Step 3

Path:

$$A \to B \to D \to F$$

$$g(F) = 1 + 3 + 2 = 6$$

$$h(F) = 1$$

$$f(F) = 6 + 1 = \boxed{7}$$

A -→ B → C → E → G ?

A* calculates:

$f(n) = g(n) + h(n)$

where

$g(n)$ = Actual cost from Start to the current node.

$h(n)$ = Estimated cost from the current node to the Goal.

$f(n)$ = Total estimated cost.

The algorithm always chooses the node with the smallest $f(n)$.

Final Shortest Path

A → C → E → G

Total Cost

$f(n) = 2 + 1 + 3 = 6$

Example 2

Why are D and F blank?

D is reached only through B.

Since B has f(B)=7, while C, E, and G all have f=6,

A* always chooses those first.

Once G is reached, the algorithm stops.

Therefore, D and F are never expanded, so their g(n) and f(n) are not computed.

AI Applications of A* (Informed Search)

1. GPS Navigation (Google Maps)

2. Autonomous Vehicles (Self-driving car)

THANK YOU

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